

Question: Explain the wireless security and MAC filtering options available, and justify which configuration approach best prevents unauthorized access while still allowing convenient onboarding of new employee devices.

Wireless Security & MAC Filtering — Easy Explanation

In this scenario, the startup has an **open-plan office** where unauthorized devices have connected before

Only employees should be able to access the Wi-Fi, while adding a new employee device should remain easy.

1. Wireless Security Options

Security	How it works	Security level
Open	No password	✗ Very poor
WEP	Old encryption method	✗ Insecure/obsolete
WPA/WPA2-Personal	Shared Wi-Fi password	☑ Good
WPA3-Personal	Modern password-based security	☑☑ Strong
WPA2/WPA3-Enterprise	Individual employee authentication through authentication server	☑☑☑ Best for organizations

For a startup, **WPA3-Personal** is a strong choice if all employee devices support it. If compatibility is an issue, **WPA2/WPA3 mixed mode** can be considered.

Every network device has a MAC address.

Example:

Employee Laptop
MAC: AA:BB:CC:11:22:33

The access point can maintain a list of allowed MAC addresses.

Allow-list

Allowed:
AA:BB:CC:11:22:33 → Laptop 1
AA:BB:CC:44:55:66 → Laptop 2
AA:BB:CC:77:88:99 → Phone

Only devices on the list are allowed to connect.

Deny-list

You can also specifically block known devices.

Blocked:

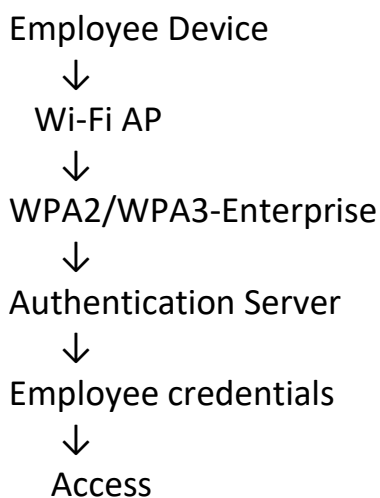
XX:XX:XX:XX:XX:XX

Best Configuration for This Scenario

For a startup that wants **security + convenient employee onboarding**, I would use:

WPA2/WPA3-Enterprise + individual employee authentication

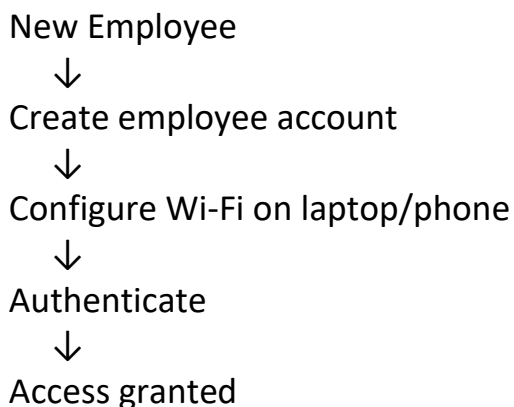
Ideally:



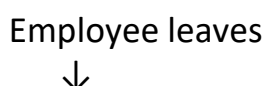
Each employee can have their **own credentials**, rather than everyone sharing one Wi-Fi password.

Why this is convenient

When a new employee joins:



When an employee leaves:



Disable their account



Their Wi-Fi access stops

You don't need to change the Wi-Fi password for the entire company.

If the startup specifically wants MAC filtering, use it as an **additional layer**, preferably with an **allow-list**, rather than relying on it as the main security mechanism.

However, maintaining a MAC allow-list for every employee device can make onboarding and device replacement more inconvenient.

So the practical priority is:

Strong authentication first → MAC filtering as optional additional control.

“Open Wi-Fi and WEP should be avoided because they provide inadequate security. WPA2/WPA3 provides encrypted access, while WPA2/WPA3-Enterprise allows each employee to authenticate individually. MAC filtering can allow or block specific devices, but it should not be relied upon as the primary security mechanism because MAC addresses can be spoofed. For a startup, WPA2/WPA3-Enterprise with individual employee authentication provides strong security and convenient onboarding because new employees can be added without changing a shared Wi-Fi password.”